HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and enhanced quantitative pricing models for energy contracts across Ireland, GB, and Europe, supporting origination and trading teams in valuing opportunities and managing risk.
- Constructed analytical frameworks to support portfolio construction by evaluating contract terms and market dynamics, contributing to strategic decision-making.
- Performed quantitative analysis on historical energy market data to identify pricing trends and potential arbitrage opportunities, informing risk management strategies.
- Collaborated with cross-functional teams, including Origination and Trading, to gather requirements for pricing model enhancements and understand market sensitivities.
- Implemented data validation checks and reconciliation processes for pricing model inputs and outputs, ensuring data integrity and accuracy.
- Designed and maintained interactive dashboards visualizing key energy market metrics and pricing model performance for senior management.
- Containerized pricing model components using **Docker** for consistent deployment across development and production environments on **AWS EC2**.

SKILLS

Core Skills – Pricing, Structuring, Quantitative Analysis, Risk Management, Portfolio Construction, Energy Markets, Contract Valuation, Numerical Methods, Data Analysis, Financial Modelling.

Programming & Scripting – Python, R, SQL.

Data Visualization & BI – Power BI, Tableau, DAX, Power Query.

Data Science & ML – pandas, NumPy, scikit-learn, Statistical Analysis, Regression, Hypothesis Testing, Machine Learning Fundamentals.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git.

Databases – PostgreSQL, MySQL, Oracle.

Enterprise Systems – ERP Integration Support, Change Management.

PROJECTS

Energy Contract Valuation Engine

- Developed a Python-based engine for valuing complex energy contracts, incorporating stochastic modelling for forward price curves and volatility.
- Integrated real-time market data feeds and historical datasets to provide dynamic pricing inputs, improving model responsiveness by **~30%**.
- Built a REST API using **FastAPI** to serve pricing results, enabling seamless integration with existing trading platforms and generating approximately **500 pricing requests** per day.

Energy Market Risk Dashboard

- Designed and implemented a **Power BI** dashboard to visualize key risk metrics and pricing sensitivities for a portfolio of energy contracts.
- Developed complex **DAX** measures to calculate Value-at-Risk (VaR) and Expected Shortfall (ES) based on market simulations.
- Automated data extraction and transformation using **Power Query** from multiple relational databases (**PostgreSQL**, **Oracle**), reducing manual reporting effort by **~15 hours** per week.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)